Majid Mahmood

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Summary

Electrical engineer with hands-on experience in the maintenance and testing of power transformers and circuit breakers, and in sizing and supervising the installation of solar PV systems. Research background in inverter-based grids, focused on how renewable generation changes short-circuit currents and the behaviour of protection devices in medium-voltage networks, including the design of an adaptive directional overcurrent protection scheme. Strong interest in protection devices and in making grids with a high share of renewable generation safe and reliable.

Research Interests

- **Protection:** short-circuit behaviour of inverter-based generation, its influence on conventional protection devices, and how grid protection can be upgraded.
- **Critical loads:** secure, uninterrupted supply for fast-growing, high-priority loads such as data centers, where faults, disturbances or protection failures in renewable-rich grids can interrupt operation.
- **Policy:** restructuring grid-connection and approval procedures, since the demand of data centers is growing faster than new grid capacity can be approved and built, so that renewable generation and new loads can connect faster without compromising security.

Education

University of Pavia Pavia, Italy
Master of Science in Electrical Engineering | Expected Grade: 105/110 *2024 - (Expected) Dec. 2026*
Courses: Electric Power Systems; Power Electronics; Energy Management; Energy Markets and Supply Structures; Systems and Technologies for Smart Grids

Khwaja Fareed University of Engineering and Information Technology Rahim Yar Khan, Pakistan
Bachelor of Science in Electrical Engineering | CGPA: 2.77/4.00 *Sep. 2018 - July 2022*
Courses: High Voltage Engineering; Power Electronics; Power System Analysis; Electronic Devices and Circuits; Transmission and Distribution

Research Experience

Master Thesis: Impact of Distributed Energy Resources on Short-Circuit Behaviour and Protection Coordination in a Modified CIGRE MV Network: An Adaptive Directional Overcurrent Approach (Expected) Dec. 2026
University of Pavia *Pavia, Italy*
Supervisor: Prof. Alessandro Bosisio

- Modelled a modified CIGRE 20 kV MV network in Python/pandapower with eight PV units and one CHP unit, and calculated short-circuit currents per IEC 60909.
- Assessed DER hosting capacity: voltage constraints give a feasible window of 31%–172% penetration, and 150% was selected as the study point.
- Showed that DER raises fault current by 9.11%–19.01% on Feeder 1, and that non-directional relays lose selectivity from 1% penetration and fail at every tested fault location by 136%.
- Coordinated IDMT/DTOC overcurrent relays (IEC 60255-151) with a 0.35 s interval and designed directional elements with torque angles derived from line impedance.
- Developed an adaptive scheme that switches relay setting groups at 31% PV via IEC 61850 GOOSE (latency about 6.1 ms), blocking reverse back-feed at 150% while keeping coordination.

Academic Projects

Power Electronics Course Project: Design and simulation of a grid connected Three-phase full-bridge inverter Mar. 2025 - Sep. 2025

University of Pavia *Pavia, Italy*
Supervisor: Prof. Pericle Zanchetta

- Designed and simulated an 8 kW three-phase grid-connected full-bridge inverter for renewable energy applications using PLECS.
- Developed a switching PWM-based inverter model composed of three independent half-bridges to generate three-phase AC voltages from a DC source.
- Designed the grid interface inductor (5 mH, current ripple 1.33 A against a limit of 2 A) and the DC-link capacitor (DC voltage ripple ≤ 0.5 V) for an 800 V DC link, 230 V grid and 10 kHz switching.
- Derived the converter transfer function and implemented PI-based current control (damping factor 0.8, natural frequency 500 Hz) for stable grid current regulation.
- Implemented DC-link voltage control (damping factor 0.85, natural frequency 10 Hz) that regulates the DC link to 800 V under varying power injection.
- Validated the system through time-domain simulations and transient power step tests, demonstrating stable active power injection and compliance with grid performance constraints.

Undergraduate Final Year Project (group of three): Centralized Control for Islanded DC Microgrid Sep. 2021 - Jun. 2022

Khwaja Fareed University of Engineering and Information Technology *Rahim Yar Khan, Pakistan*
Supervisor: Engr. Dr. Muhammad Umair Shahid

- Designed the three DC-DC buck converters (48 V input, regulated 12 V output, modules rated up to about 230 W each) supplying an LED light, a DC motor and a fan, with my supervisor's guidance.
- Derived a mathematical model of the converter and calculated and tuned the PID gains for the voltage feedback.
- Simulated the converters and control in MATLAB/Simulink.
- Wrote the project thesis (aligned with SDGs 7, 9 and 13). The group prototype held output voltages of 11.99 V, 12.06 V and 12.21 V, with load sharing between the converters.

Professional Experience

Freelance Field Engineer Dec. 2025 - Present

TNS IT Solutions *Milan, Italy*

- Installed DC power feeds for network equipment in a Milan data center (-48 V, 63 A rectifier breakers to the equipment power modules) and a residential inverter with battery backup.

Junior Electrical Engineer Nov. 2022 - Dec. 2023

Lahore Electric Supply Company-LESCO *Lahore, Pakistan*

- Managed the cleaning of power contacts of SF6 Circuit Breakers by using high-pressure water jets to remove dirt.
- Systematically tested Power Transformers under various conditions to improve safety standards and document results.
- Assisted in the inspection of control schemes of Circuit Breakers and Magnetic Contactors.
- Record, analyze, and report the data to enhance the reliability and safety of power transformers.

Solarization of Public sector Buildings-LESCO *Lahore, Pakistan*

- Conducted surveys to assess the feasibility of implementing renewable energy in government offices.
- Calculated the appropriate size of photovoltaic (PV) systems based on the energy loads and consumption patterns of government offices.
- Oversaw installation teams to ensure compliance with design specifications and electrical safety standards.
- Provided technical support during and after installation for optimal system performance.

Engineering Intern Jul. 2022 - Sep. 2022

Dehkan Flour Mills (Pvt) Ltd *Rahim Yar Khan, Pakistan*

- Performed basic maintenance and troubleshooting on motors, conveyors, control panels, and industrial electrical systems.
- Assisted in electrical engineering projects related to automation and energy-efficiency improvements.

Technical Skills

Power System Analysis & Simulation: pandapower, MATLAB, PowerWorld

Power Electronics Simulation: PLECS, Proteus

Engineering & GIS Tools: AutoCAD, QGIS

Programming: Python, C++

Research & Documentation: Technical Writing, MS Office, LaTex

Awards & Honors

- **EDISU Scholarship Recipient (2026-Italy):** Awarded a competitive regional scholarship for Master's studies in Electrical Engineering at the University of Pavia, based on academic merit and eligibility criteria established by the Regione Lombardia.
- **Prime Minister's Laptop Scheme Award (2017-Pakistan):** Received a merit-based laptop award under the government initiative for outstanding academic performance in Matriculation examinations.

SCIENTIFIC COMMUNICATION & ENGAGEMENT

- **Scientific Writing:** Experienced in drafting technical reports and comprehensive thesis chapters.
- **Presentation & Dissemination:** Skilled in delivering technical presentations at group meetings and adept at communicating complex electromechanical data to multidisciplinary audiences.
- **Collaborative Research:** Active participant in academic discussions and group meetings.

Professional Memberships

- **Registered Electrical Engineer** 2022
  Pakistan Engineering Council (PEC)

Training Certificates

- **Workplace Safety: General and Specific Low Risk Training (D.Lgs. 81/08)** Oct. 2024
  University of Pavia
- **Industrial Automation (PLC & HMI)** Jan. – Mar. 2023
  PITAC (Pakistan Industrial Technical Assistance Centre)
- **Solar PV Design & Installation** Dec. 2022 – Jan. 2023
  PITAC (Pakistan Industrial Technical Assistance Centre)
- **AutoCAD with Switchgear Panel Designing** Dec. 2022 – Jan. 2023
  PITAC (Pakistan Industrial Technical Assistance Centre)
- **Industrial Electrician** Nov. 2022 – Jan. 2023
  PITAC (Pakistan Industrial Technical Assistance Centre)
- **Introduction to the Internet of Things and Embedded Systems** Aug. 2022
  University of California, Irvine (Coursera)
- **International Training Workshop on High Voltage Laboratory** 14–15 Jul. 2022
  Khwaja Fareed University of Engineering & Information Technology
- **English Proficiency** 2022
  Khwaja Fareed University of Engineering & Information Technology

Languages

- **English:** Fluent
- **Italian:** Intermediate
- **Urdu:** Native